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Quantum Hybrid Models for Breast Cancer Screening

Emiliano Lorenzo

Matricola 7177126

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1 Introduction

This project is focused on the utilization of Quantum Machine Learning in the medical field. In particular, the context of this research is Breast Cancer Screening. Breast cancer is a highly prevalent pathology and one of the leading causes of cancer-related mortality among women worldwide [1], making early diagnosis and prevention crucial clinical priorities. With this aim, this project focused on applying Artificial Intelligence, a technology that recently gained strong popularity thanks to its capacities of analysis of big data of every type, to early diagnosis of Breast Cancer. This work is based on the article "Quantum integration in swin transformer mitigates overfitting in breast cancer screening" by Zongyu Xie et al. [4]. In clinical practice, X-ray digital mammography and ultrasonography are the primary modalities employed to detect breast cancer in its early stages [2] and this reference study utilized Full-Field Digital Mammography (FFDM) for its experiments. This project intentionally adopts ultrasound imaging data to validate the generalizability of quantum-enhanced models across different biomedical imaging modalities with respect to the ones analyzed in the reference paper. The other corner stone of this study is the utilization of Quantum Machine Learning techniques. Quantum Machine Learning is an emerging field of study that combines data driven artificial intelligence (machine learning) and quantum computing. Driven by recent hardware advancements, such as the realization of physical quantum processors exceeding one hundred qubits [3], the scientific community has increasingly focused on leveraging quantum computing to enhance machine learning operations. In the medical field, and in particular in the digital image analysis, object of this research, Quantum Machine Learning can provide strong improvements to the performance of models in data analysis. The reference paper introduces utilized models in detail, with a particular focus on the overfitting mitigation achieved by quantum layers compared to classical deep learning benchmarks. Starting from this strong baseline, this project aims to reproduce the reported results—despite the unavailability of the authors' proprietary dataset, source code, and specific quantum hardware—and subsequently focuses on exploring further possibilities considering the paper achievements as a baseline. Reproducibility was another important objective of this research. With this aim, all the code has been written using well documented frameworks, such as PyTorch [5] for Deep Learning and PennyLane for Quantum Computing [6], as well as publicly available dataset like BreastMNIST, part of the MedMNIST collection [7]. During the training phase, the code was executed in simulation mode using PennyLane backends. Ultimately, to assess the stability and hardware portability of the learned representations, the best performing architecture was tested on a physical quantum processor via the IBM Quantum Platform [8]. The results not only confirmed the findings of the reference paper but, in several instances, outperformed them, offering the additional advantage of full reproducibility facilitated by the employment of open-source resources.

2 Datasets

During this study two main datasets were considered: INbreast and BreastMNIST [9] [7]. This choice was driven by several major reasons. First of all, the reference paper [4] used for its analysis a proprietary dataset comprising Full-Field Digital Mammography (FFDM) examinations collected from patients in China between January 2019 and August 2021. Although the original authors were contacted well in advance to request access, no response was received, rendering this primary dataset unavailable for reproduction. This

dataset was designated as Cohort A and utilized in the reference study as the internal cohort for training and validation. For external validation, the authors employed the INbreast dataset [9], referred to as Cohort B, to assess the generalization capabilities of their models. INBreast is a publicly available dataset, so it was first considered as the main candidate for this project. Unfortunately, due to technical limitations detailed in subsequent sections, INbreast did not satisfy the strict requirements for training the complex architectures involved in this project. Consequently, BreastMNIST was selected as a robust alternative, also with the aim of validating the robustness of the original paper results on a different benchmark.

2.0.1 Dataset: INbreast

The INBreast database is a publicly available dataset released by the Breast Centre in CHSJ, Porto between April 2008 to July 2010. Other releases of this dataset are also available on Kaggle, which were the ones considered for this study. As previously mentioned, in the reference paper INbreast was designated as Cohort B and exclusively used for external evaluation. Specifically, only 107 cases with Region of Interest (ROI) annotations were included, while all others were excluded. ROIs represent specific lesion areas manually annotated by radiologists. The images contained in this dataset are in DICOM format (digital imaging and communications in medicine), and were converted in Portable Network Graphics (PNG) format by the authors of the paper using min-max normalization.

The data preprocessing for the INbreast dataset in this study was structured as an automated Python pipeline for data ingestion, clinical mapping, and spatial transformation. The objective of the analysis implemented in the code was primarily to extract localized Regions of Interest (ROIs) and format them as standardized inputs (224×224 pixels) suitable for the Swin Transformer and QEST architectures.

- **Data Ingestion and Metadata Extraction**
  The pipeline initiates with the extraction of INbreast dataset from a compressed archive into a volatile, high-speed local directory. The script bypasses metadata files (e.g., .xls, .xml, .roi) and isolates .dcm (DICOM) files. To optimize memory and computational overhead, the pydicom library is utilized with the `stop_before_pixels=True` parameter. This allows the system to parse only the essential DICOM headers (such as PatientID, ImageLaterality, and spatial dimensions) without loading the heavy pixel arrays into RAM.

- **Ground Truth Integration and Label Binarization**
  DICOM metadata are linked with the official INbreast diagnostic spreadsheet (Excel/CSV) to establish ground truth. Specifically, a Left-Join operation merges the imaging dataframe with the clinical dataframe using the `PatientID` and `File Name` keys, while a custom mapping function converts standard BI-RADS (Breast Imaging-Reporting and Data System) categories into a binary classification problem. Following standard clinical thresholds, BI-RADS categories 1, 2, and 3 are aggregated into the "Benign" (Class 0) cohort, while categories 4, 5, and 6 are classified as "Malignant" (Class 1).

- **Spatial Parsing and Image Cropping (Function Analysis)**
  This is the core of the preprocessing pipeline, which defines specific functions to extract the lesion areas. parse_roi_file(roi_path) utilizes Regular Expressions (Regex)

to parse these files and extract the explicit mathematical coordinates (e.g., {x, y}) that delineate the lesion's contour. get_bounding_box(points, img_shape) generates bounding boxes with padding for neural networks. process_single_case(dicom_path, roi_path_list) serves as a rigorous filtering heuristic to isolate the most clinically relevant lesion when multiple annotations exist.

Finally, the script executes a sanity check by sampling random malignant cases, applying the entire extraction pipeline, and plotting the original high-resolution FFDM alongside the 224×224 processed crop.

During the Exploratory Data Analysis (EDA), several critical issues emerged. First, the ROI annotations were generated using the OsiriX software, which proved difficult to parse. Even when utilizing the XML format, the extracted areas often lacked sufficient detail to serve as meaningful inputs for a neural network, primarily due to the predominance of dark background pixels and the difficulty of isolating the tumor masses. Furthermore, manual annotations by radiologists resulted in highly heterogeneous bounding boxes, many of which encompassed pectoral muscles rather than isolating the tumors. Filtering out these problematic images led to a severe reduction in dataset size, rendering it insufficient for advanced model training and validation.

2.0.2 Dataset: BreastMNIST

To overcome the severe data attrition, annotation noise, and computational bottlenecks encountered during the Exploratory Data Analysis (EDA) of the INbreast dataset, the BreastMNIST dataset was adopted as a robust and standardized alternative for model training and evaluation. This dataset belongs to the MedMNIST collection and contains 780 images, which are pre-processed, standardized, and rigorously partitioned by the original authors into a 7:1:2 ratio: 546 images for training, 78 for validation, and 156 for testing. This predefined split ensures a fair and reproducible benchmarking environment. Unlike INbreast, consisting of Full-Field Digital Mammography (FFDM) X-rays, BreastMNIST is comprised exclusively of breast ultrasound images. While this introduces a shift in the imaging modality—from measuring tissue radiodensity to acoustic echogenicity—it provides a highly controlled environment to purely isolate and evaluate the architectural details tested. Moreover, it is an opportunity to assess the robustness of the solution proposed in the original paper to data format shift.

To rigorously assess the morphological and statistical properties of the BreastMNIST dataset, an automated Exploratory Data Analysis (EDA) pipeline was developed in Python also in this case. Matplotlib and Seaborn were used as libraries for plots. The script executes the following sequential steps to extract and visualize key dataset characteristics:

- **Class Distribution Analysis:** The class balance of the dataset was evaluated, revealing a significant imbalance: the benign class was overrepresented compared to the malignant class by a ratio of approximately 2.7 to 1. Consequently, the `class_weights_classic_1` variable was configured to penalize misclassifications of malignant cases roughly three times more heavily. Subsequent experiments revealed this initial adjustment to be insufficient; thus, to further enhance model performance, a `class_weights_aggressive` variable was instantiated. This applied a stricter 4:1 weighting ratio in favor of malignant cases, ultimately yielding superior classification results.

- **Morphological Sample Visualization:** To facilitate qualitative visual inspection, the script extracts the first five samples from both the malignant and benign cohorts within the training set. These samples are plotted in a 2×5 grid. The images are rendered in grayscale at their native 28×28 pixel resolution, allowing for a preliminary assessment of the ultrasound textures associated with each diagnostic class.

- **Pixel Intensity Profiling:** A statistical evaluation of the image brightness is performed by flattening the training images into a 1D array and normalizing the pixel values to a $[0, 1]$ continuous scale. A histogram overlaid with a Kernel Density Estimation (KDE) curve is plotted to illustrate the probability density of pixel intensities across the dataset. The global mean pixel intensity is also calculated and superimposed on the plot to identify the overall radiometric baseline.

- **Spatial and Statistical Pattern Extraction:** To identify distinct positional features between the two classes, the script computes the "mean image" for both the malignant and benign sets by averaging the pixel matrices across all respective samples. These average images are visualized using a high-contrast colormap (`magma`). Finally, a spatial difference map is computed by subtracting the average benign image from the average malignant image. Visualized with a diverging colormap (`coolwarm`), this difference map highlights specific spatial regions where malignant tumors exhibit systematically higher or lower echogenicity compared to benign samples.

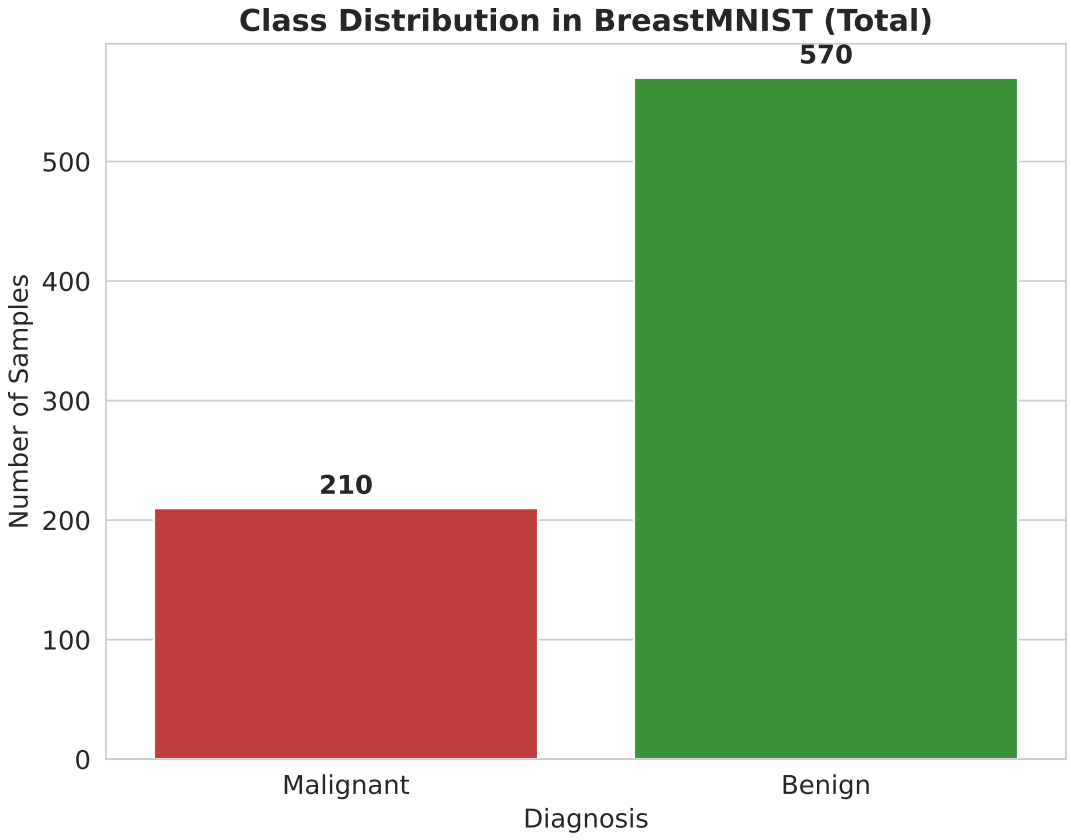


Figure 1: Class Distribution of BreastMNIST

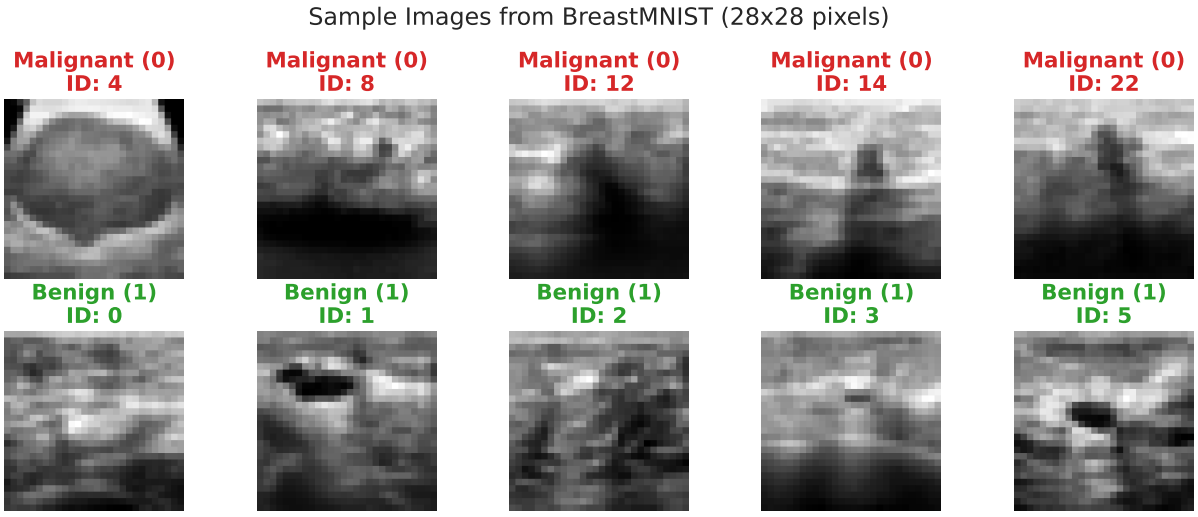


Figure 2: Grid of some samples

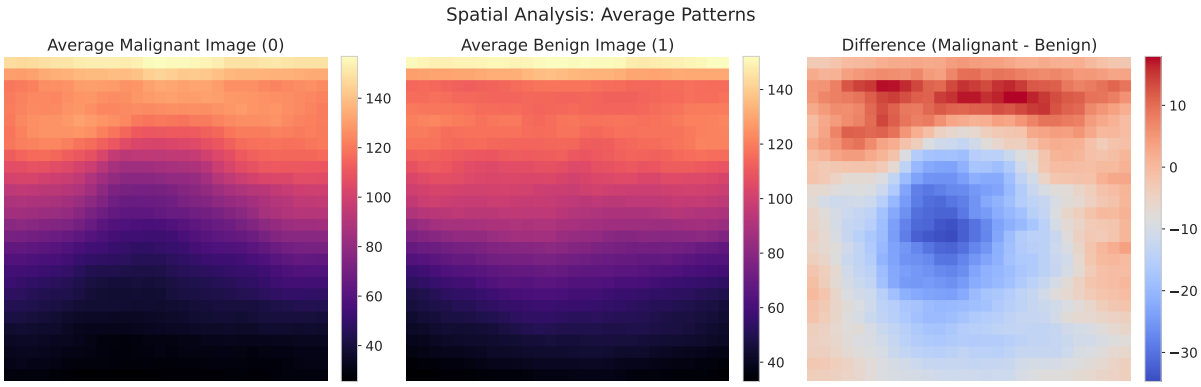


Figure 3: Average Patterns in BreastMNIST

3 Model Architectures

This section details the two architectures compared in the study. Following the methodology of the reference paper [4], two main categories were evaluated: classical deep learning models and quantum-hybrid models. Currently, deep learning architectures, such as large-scale transformer-based networks and Convolutional Neural Networks (CNNs), achieve state-of-the-art performance on image classification tasks. Consequently, it was a natural choice to follow the reference study in adopting such models as a robust baseline. As previously introduced, the primary novelty of the original research lay in the direct comparison between classical deep learning models and quantum-hybrid architectures [16, 11]. This comparative approach was replicated in the present research, with the addition of a deep architectural exploration of quantum-hybrid configurations to establish whether the reported performance could be further improved. For the classical baseline, the Swin Transformer [12] was adopted as the standard state-of-the-art architecture. Regarding the quantum machine learning domain, the evaluated models belong to the quantum-hybrid context, perfectly aligning with the framework proposed in the original paper.

3.1 Classic Deep Learning: Swin Transformer

Swin Transformer was introduced in 2021 from Microsoft Research Asia [12]. It is a direct descendent of Vision Transformer (ViT) architecture. Standard ViTs divide images in patches (usually 16 x 16) and compute Self Attention globally among all the patches. This implementation leads to quadratic complexity with respect to the image resolution. In medical field usually this computational cost is a limitation because of the high resolution images models have to deal with. Swin Transformers overcome these issues computing attention in local non overlapping windows. In every block, the network applies Window Multi-Head Self-Attention (W-MSA) limiting the focus on a single window (e. g. 7 x 7 patch). In the following block Shifted Window (SW-MSA), a shifting mechanism enabling cross-window communication, shifts the windows by half their size. This implementation allows cross-window communication, effectively propagating global information without incurring a quadratic computational cost. Between the stages, group of near patches are fused together, reducing spatial resolution (something similar happens with pooling in CNNs) but augmenting channel depth. As explained, thanks to local windows the computational cost scales linearly with respect to the input dimension. Another fundamental advantage, especially in the medical images field, deals with the Multiscale Capacity. Traditional ViTs never change feature scale, while Swin Transformers, thanks to their hierarchical structure, can both extract finest details (local features of the first steps) and larger semantic elements, as global features in the last stages.

3.1.1 Architectural Details

As in the original paper [4], also in this study the Swin Transformer model is downloaded as a pre-trained backbone and then fine tuned on the specific dataset. This is a common procedure in all Deep Learning. Swin Transformers have tens of millions of parameters and it is not possible to train a model such that completely from scratch on a relatively small dataset like the ones considered in the original paper or in this project. For coherence with the paper, Swin-B Transformer from `torchvision.models`, pretrained on ImageNet 1K [13] was selected. This model has 88 million parameters. Given that the BreastMNIST dataset comprises single-channel grayscale ultrasound images, a structural adaptation of the initial patch embedding layer was strictly required. The original three-channel (RGB) 2D convolutional layer was replaced by a functionally equivalent single-channel convolution. To rigorously preserve the hierarchical spatial filters learned during the ImageNet pre-training phase, the new kernel weights were deterministically initialized by summing the pre-trained weights across the channel dimension, while the original bias terms were fully retained. In addition to these implementation details, to allow full fine tuning of the model the `requires_grad` attribute was set to True for all parameters, unfreezing the entire network and allowing the hierarchical self-attention mechanisms to optimally recalibrate to the specific radiometric and morphological properties of breast lesions. Furthermore, the default classification head of the Swin-B model was discarded and replaced with a custom Multi-Layer Perceptron (MLP). The feature vector extracted by the Swin-B backbone, consisting of 1024 dimensions, is first passed through a linear layer of equal dimensionality (1024 to 1024), followed by a non-linear ReLU activation function. Then, a subsequent linear bottleneck layer is used to severely compress the feature space from 1024 down to a reduced number of dimensions (8 or 16). This specific size of dimensions is designed to mirror the bottleneck used in the quantum architectures that will be further illustrated, to achieve a fair comparison. Finally, a terminal

linear layer maps the bottleneck vector to the two target diagnostic classes (Benign and Malignant). This tailored architectural configuration ensures a perfectly equitable and methodologically sound comparison between the classical fully connected classifier and its quantum-enhanced counterpart, isolating the structural impact of the quantum layers on the model's predictive performance and overfitting mitigation capabilities.

3.2 Quantum Machine Learning

Quantum Machine Learning (QML) is an interdisciplinary field situated at the intersection of classical machine learning and quantum computing. The convergence of these two research areas is driven by recent technological advancements and the inherent need within classical machine learning for highly expressive embedding and efficient sampling techniques [14]. For instance, kernel methods heavily rely on complex data embeddings into high-dimensional feature spaces [15], while generative AI fundamentally depends on representing and sampling from intractable probability distributions. Exploiting the principles of quantum mechanics, quantum computing offers theoretical and practical advantages in both contexts, making the integration of these disciplines a natural scientific evolution.

Within the QML domain, contemporary research frequently focuses on quantum-hybrid architectures. These models seamlessly integrate classical neural network layers with parameterized quantum circuits [16]. A primary motivation for this hybrid approach is the severe bottleneck associated with quantum data encoding, also known as state preparation. Classical data must be embedded into a quantum Hilbert space to be processed by quantum hardware, and this embedding phase often constitutes the most critical computational limitation. To mitigate this issue, classical deep learning models are widely employed as feature extractors. By performing drastic dimensionality reduction on the raw input, they facilitate a much more efficient and scalable encoding of the latent representations into the subsequent quantum circuits. Several types of encoding have been studied, the principal ones are:

- **Angle Encoding:** This technique is able to encode discrete and continuous features into rotation angles (e.g., via R_x, R_y, or R_z gates). Angle Encoding allows hardware-efficient design, making it highly suitable for implementation on real quantum computers operating within the Noisy Intermediate-Scale Quantum (NISQ) era.

- **Amplitude Encoding:** This method encodes an N-dimensional classical vector directly into the probability amplitudes of a quantum state using n qubits, where $n = \lceil \log_2 N \rceil$. While it provides an exponential advantage in data compression (capacity), it requires preliminary L^2-normalization of the classical data and inherently demands non-trivial quantum circuits. Consequently, it is heavily susceptible to noise on current physical hardware.

- **Basis Encoding:** In this approach, classical binary strings are mapped directly to the computational basis states of a quantum system (e.g., a classical "011" becomes the quantum state $|011\rangle$). It requires N qubits to encode N binary features. Although theoretically straightforward, it only supports binary numbers, making it fundamentally unsuitable for continuous, real-valued feature vectors.

- **Arbitrary Encoding:** A more generalized state preparation paradigm designed to map complex, arbitrary classical data distributions into generic quantum states.

While it theoretically allows for maximum representational flexibility, synthesizing an arbitrary quantum state mathematically demands high gate complexity and prohibitive circuit depths. As a result, its practical applicability remains severely limited in contemporary quantum machine learning workflows.

In the context of the original paper [4], Angle Encoding, Amplitude Encoding and Basis Encoding were examined, identifying Angle Encoding as the only feasible technique for the discussed application. Due to this fact and to the previously exposed alternatives, also this project focused only on Angle Embedding, after considering the strong results it allowed to achieve during the experimental phase. The Angle Embedding was implemented as follows:

- **Non-linear Normalization (Preprocessing):** To map the unconstrained classical feature values into the valid domain of quantum gate rotation angles $[-\pi/2, \pi/2]$, a non-linear normalization is applied using the arctangent function. Mathematically, each element x_i of the input feature vector is transformed as:

$$x_i' = \arctan(x_i) \tag{1}$$

  In the Python pipeline, this is efficiently vectorized using the `torch.atan(inputs)` operation.

- **Superposition Initialization:** Following standard quantum state preparation, each qubit is initialized in the zero state $|0\rangle$ and subsequently placed into a uniform superposition. This is achieved by applying a Hadamard gate to each i-th wire:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \tag{2}$$

  This corresponds to the `qml.Hadamard(wires=i)` instruction in the loop.

- **Feature Encoding via Orthogonal Rotations:** The normalized classical values are embedded into the quantum state through a sequence of rotation gates. Specifically, x_i' is mapped to the angle of a Y-rotation (R_y), while its square $(x_i')^2$ is mapped to the angle of a Z-rotation (R_z) to enhance the representational non-linearity of the circuit. The combined operation on the state is represented as:

$$|\psi_{x_i}\rangle = R_z((x_i')^2)R_y(x_i')\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) \tag{3}$$

  In the PennyLane framework, this is executed sequentially via `qml.RY(angle, wires=i)` and `qml.RZ(angle**2, wires=i)`. The overall multi-qubit embedded state $|\psi_x\rangle$ is the tensor product of all individual qubit states, formulated as:

$$|\psi_x\rangle = \bigotimes_{i=1}^{n} |\psi_{x_i}\rangle = \bigotimes_{i=1}^{n} R_z((x_i')^2)R_y(x_i')\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) \tag{4}$$

The other corner stone of Quantum Machine Learning are Variational Quantum Circuits (VQC). These architectures are the quantum equivalent of classic Deep Neural Networks because of the fact that both are composed of trainable parameters. While in the first the parameters are software variables, in the quantum version parameters are directly mapped to the hardware. Usually rotation amplitudes are left as trainable parameters that can change during training. Following the original paper, in this project architectures inspired by the circuit-centric quantum classifier were adopted [17].

3.2.1 Quantum Enhanced Swin Transformer

The novel element introduced by the paper [4] is Quantum Enhanced Swin Transformer (QEST). This model is based on the quantum transfer learning paradigm [16], where classical machine learning backbones are used as feature extractors for quantum layers, but it is introduced for the first time by the authors [4] (at the best of their knowledge) and specifically for the Breast Cancer screening task. In this project, the hybrid architecture proposed by the original study serves as a foundational baseline for the subsequent exploration of alternative quantum-hybrid implementations. The following sections detail the specific architectural choices and design configurations adopted for the quantum models developed in this research.

3.2.2 Variational Quantum Circuit (VQC) Architecture

In this section, the investigated architectures are detailed. Each of these implementations explores different ansatz designs, aiming to compare the effectiveness of various configurations for the task at hand. These variations were evaluated to determine whether any performance improvements could be achieved. Every described circuit is encapsulated within a `QNode` using PennyLane's built-in Python decorators. This represents the standard procedure for integrating quantum variational layers into classical deep learning architectures. Specifically, this approach leverages PyTorch's capability to convert PennyLane `QNode`s into compatible neural network layers, seamlessly embedding them into the computational graph to enable the automatic differentiation essential for model training. Finally, the architectures detailed below are referred to by the names of their corresponding Python functions as implemented in the project's source code.

Python function name: `circuit`

The first implementation of VQC faithfully replicates the architecture presented in the original paper. This was done with the aim of constituting a strong baseline to use as a benchmark for further developments. The Python function implementing this in the project is `circuit`. According to the paper, following the quantum embedding phase, the classical feature representations are processed by a parameterized Variational Quantum Circuit (VQC). In the provided computational framework, this architecture is implemented via the `circuit` function, optimized as a PennyLane `QNode`. The VQC is structured into L identical consecutive layers (with L determined dynamically by the first dimension of the `weights` tensor), where each layer consists of a variational learning module and an entanglement module, followed by a final measurement readout.

- **Variational Learning Module:** To introduce trainable parameters into the quantum network, a universal single-qubit rotation gate, $U_3(\theta, \phi, \lambda)$, is applied to each qubit. The U_3 gate enables arbitrary rotations on the Bloch sphere, optimizing the quantum state representation during the learning phase. The operation is mathematically defined by the following matrix:

$$U_3(\theta, \phi, \lambda) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -e^{i\lambda}\sin\left(\frac{\theta}{2}\right) \\ e^{i\phi}\sin\left(\frac{\theta}{2}\right) & e^{i(\phi+\lambda)}\cos\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{5}$$

  In the Python script, this sub-phase is executed through the `qml.U3(theta, phi, lam, wires=q)` instruction, iteratively extracting the three parameter values from the weight tensor and applying the rotation across all n qubits.

- **Entanglement Module:** To capture complex correlations among the encoded features, spatial entanglement is generated using Controlled-Z (CZ) gates. The architecture adopts a strict entanglement topology, mapping the CZ gates sequentially from qubit q to the adjacent qubit $q+1$. This linear chain is programmatically implemented via the `qml.CZ(wires=[q, q + 1])` function. Notably, this topology was deliberately selected in the original paper to perfectly align with the physical connectivity constraints of the target quantum hardware, thereby circumventing the need for computationally expensive SWAP gates during circuit compilation.

- **Measurement Module:** The final quantum state, denoted as $|\psi_y\rangle$, is mapped back to the classical domain by computing the expectation values of the Pauli-Z observables across all N qubits. This collective measurement yields the continuous output feature vector $\vec{y}$, which is subsequently passed to the classical terminal classifier:

$$\vec{y} = \langle\psi_y|Z^{\otimes N}|\psi_y\rangle \tag{6}$$

  In the PennyLane Python script, this global measurement vector is programmatically constructed using the `return` statement. Specifically, a list comprehension evaluates the local `qml.PauliZ` observable for each individual qubit, aggregating them into the final structured array corresponding to $\vec{y}$: `[qml.expval(qml.PauliZ(i)) for i in range(n_qubits)]`.

Python function name: `circuit_strong`

To rigorously assess the impact of higher circuit expressivity and non-local qubit connectivity, an alternative architecture was developed and tested against the baseline model. The Python function implementing this advanced ansatz is `circuit_strong`. Following the standard `custom_angle_embedding` phase, the classical representations are processed by a highly parameterized quantum neural network template. In this computational framework, the architecture is implemented leveraging the `qml.StronglyEntanglingLayers` function provided by PennyLane, optimized within a `QNode`. This template abstracts the explicit loop definitions by intrinsically applying a sequence of parameterized rotations and wide-range entangling operations across L layers, defined by the `weights` tensor of shape `(n_layers, n_qubits, 3)`. The architecture consists of a rotational learning module, a long-range entanglement module, and a final measurement readout.

- **Rotational Learning Module:** Instead of utilizing the standard U_3 gate, the `StronglyEntanglingLayers` template introduces trainable parameters through a sequence of three elementary Pauli rotations. For each layer l and qubit q, arbitrary single-qubit rotations are performed using the general rotation gate $Rot(\phi, \theta, \omega)$. This operation is mathematically equivalent to an Euler angle rotation sequence $R_z - R_y - R_z$, defined by the following unitary matrix product:

$$Rot(\phi, \theta, \omega) = R_z(\omega)R_y(\theta)R_z(\phi) \tag{7}$$

  In the Python script, PennyLane automatically extracts the three corresponding parameters from the provided `weights` tensor for each qubit and iteratively applies this composite rotation, optimizing the system's state space traversal.

- **Long-Range Entanglement Module:** Unlike the strict nearest-neighbor CZ topology employed in the baseline architecture, this module captures highly complex, non-local feature correlations. It achieves this by applying Controlled-NOT

($CNOT$) gates across varying spatial ranges. Specifically, for a given layer, the entanglement connects qubit q to a target qubit $(q+r) \pmod n$, where r is a layer-dependent hyperparameter (range) that dictates the "stride" of the entanglement. By setting `ranges=None` (default behavior) within the `qml.StronglyEntanglingLayers` function call, the framework dynamically computes the optimal strides r for each layer. While this topological ansatz significantly enhances the theoretical expressivity and entanglement capacity of the model, it is generally more susceptible to noise on current NISQ hardware due to the necessity of deep SWAP gate compilations for distant physical qubits.

- **Measurement Module:** Consistent with the baseline approach, the final quantum state is mapped back to the classical domain by computing the expectation values of the Pauli-Z observables. The continuous expectation values provide the output feature vector:
$$\vec{y} = \langle \psi_y | Z^{\otimes N} | \psi_y \rangle \tag{8}$$
This measurement is executed by the terminal `return` statement containing the list comprehension: `[qml.expval(qml.PauliZ(i)) for i in range(n_qubits)]`.

Python function name: `circuit_reupload_template`

To further investigate methods for enhancing the expressivity and non-linearity of the Variational Quantum Circuit (VQC), an architecture based on the *Data Re-uploading* paradigm [18] was formulated and evaluated. The Python function implementing this advanced ansatz is `circuit_reupload_template`. Unlike the baseline model, which encodes the classical information into the quantum state only once at the beginning of the circuit, this architecture dynamically re-injects (re-uploads) the classical feature representations at the onset of every parameterized layer. The VQC is structured into L identical consecutive layers (with L determined dynamically by the first dimension of the `weights` tensor), where each layer now consists of a data re-uploading module, a variational learning module, and an entanglement module, culminating in a final measurement readout.

- **Data Re-uploading Module:** To continuously "refresh" the quantum state with the input classical data, the entire embedding unitary $U_{emb}(x)$ is sequentially re-applied. For a given layer l, the quantum state transitions according to the sequential application of the embedding, variational (U_{var}), and entanglement (U_{ent}) unitaries:
$$|\psi_l\rangle = U_{ent}\, U_{var}(\theta_l)\, U_{emb}(x)\, |\psi_{l-1}\rangle \tag{9}$$
In the Python script, this continuous data injection is handled by invoking the `custom_angle_embedding_dynamic(inputs, n_qubits)` function at the beginning of the layer loop. This guarantees that the original features are recursively encoded via orthogonal R_y and R_z rotations, increasing the universal approximation capabilities of the model.

- **Variational Learning Module:** Following the data re-uploading, trainable parameters are introduced using the universal single-qubit rotation gate, $U_3(\theta, \phi, \lambda)$, applied to each qubit. As previously said, the U_3 gate enables arbitrary rotations on the Bloch sphere, optimizing the quantum state representation. The operation is mathematically defined by matrix previously introduced:
$$U_3(\theta, \phi, \lambda) = \begin{pmatrix} \cos(\frac{\theta}{2}) & -e^{i\lambda}\sin(\frac{\theta}{2}) \\ e^{i\phi}\sin(\frac{\theta}{2}) & e^{i(\phi+\lambda)}\cos(\frac{\theta}{2}) \end{pmatrix} \tag{10}$$

- **Entanglement Module:** To capture complex spatial correlations among the newly re-uploaded features, entanglement is generated using Controlled-Z (CZ) gates. The architecture retains the strict entanglement topology of the baseline, mapping the CZ gates sequentially from qubit q to the adjacent qubit $q+1$. This linear chain is implemented via the `qml.CZ(wires=[q, q + 1])` function, ensuring hardware-efficient physical connectivity constraints.

- **Measurement Module:** Upon the completion of the L layers, the final quantum state is mapped back to the classical domain by computing the expectation value of the Pauli-Z observable for each individual qubit, as previously explained.

Python function name: `circuit_reupload_template_stab`

To rigorously address the quantum mechanical instabilities associated with naïve data re-uploading architectures, an optimized and stable version of the circuit was developed. The Python function implementing this refined ansatz is `circuit_reupload_template_stab`. In iterative quantum neural networks, repetitively applying Hadamard gates within the layer loop acts as a basis transformation that inadvertently destroys the complex phase relations and entanglement learned in preceding layers. To preserve quantum coherence and ensure stable gradient propagation, this architecture strictly decouples the initial superposition state preparation from the parameterized data re-injection loop. The VQC is structured into a singular initialization phase followed by L dynamic layers, each comprising a stabilized data re-uploading module, a variational learning module, and an entanglement module.

- **Initial State Preparation:** Before entering the iterative layer loop, the multi-qubit system is initialized into a uniform superposition state exactly once. This is achieved by applying a Hadamard gate to each i-th wire in the ground state:

$$|\psi_0\rangle = \bigotimes_{i=0}^{n-1} H|0\rangle_i = \bigotimes_{i=0}^{n-1} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)_i \tag{11}$$

  In the Python script, this foundational step is executed via a dedicated loop running `qml.Hadamard(wires=i)` strictly outside the main architectural loop.

- **Stabilized Data Re-uploading Module:** At the onset of each layer l, the classical input features are dynamically re-encoded into the evolving quantum state $|\psi_{l-1}\rangle$ without disrupting its fundamental superposition. The transformation maps the normalized classical values exclusively to the angles of Y- and Z-rotations:

$$U_{emb}(x_i) = R_z((x_i')^2)R_y(x_i') \tag{12}$$

  Programmatically, this is handled by the auxiliary `custom_angle_embedding_dynamic_stab(inputs, n_qubits)` function, which deliberately omits the Hadamard gate to act purely as a data-injection unitary.

- **Variational Learning Module:** Following the stabilized data re-uploading, trainable parameters are applied using the universal single-qubit rotation gate $U_3(\theta, \phi, \lambda)$.

- **Entanglement Module:** To capture complex spatial correlations, spatial entanglement is sequentially generated using Controlled-Z (CZ) gates. The architecture adopts a nearest-neighbor entanglement topology.

- **Measurement Module:** The final quantum state is mapped back to the classical domain by computing the continuous expectation value of the Pauli-Z observable for each individual qubit.

Python function name: `circuit_reupload_template_stab_en`

To further augment the expressivity of the stabilized data re-uploading architecture and maximize the propagation of quantum information across the register, an enhanced entanglement topology was evaluated. The Python function implementing this variant is `circuit_reupload_template_stab_en`. While the initial state preparation, data injection, and variational learning modules remain structurally identical to the stabilized baseline, this architecture fundamentally alters the spatial correlation structure by transforming the linear qubit chain into a closed ring topology.

- **Initial State Preparation:** The multi-qubit system is uniformly initialized exactly once prior to the iterative loop by applying a Hadamard gate to each wire in the ground state:

$$|\psi_0\rangle = \bigotimes_{i=0}^{n-1} H|0\rangle_i \tag{13}$$

- **Stabilized Data Re-uploading Module:** At each layer l, the classical input features are dynamically re-encoded into the quantum state without disrupting its coherence, mapping the normalized classical values to the angles of Y- and Z-rotations:

$$U_{emb}(x_i) = R_z((x_i')^2)R_y(x_i') \tag{14}$$

- **Variational Learning Module:** Trainable parameters are applied using the universal single-qubit rotation gate $U_3(\theta, \phi, \lambda)$. For a given layer l and qubit q, the operation is defined as:

$$U_3(\theta, \phi, \lambda) = \begin{pmatrix} \cos(\frac{\theta}{2}) & -e^{i\lambda}\sin(\frac{\theta}{2}) \\ e^{i\phi}\sin(\frac{\theta}{2}) & e^{i(\phi+\lambda)}\cos(\frac{\theta}{2}) \end{pmatrix} \tag{15}$$

- **Ring Entanglement Module:** Unlike the strict linear array utilized in previous iterations, this module forces the terminal qubit to connect back to the initial qubit. This ensures that every qubit symmetrically interacts with exactly two neighbors, preventing potential boundary bottlenecks in information flow. In the Python script, this is explicitly implemented by appending the conditional instruction `if n_qubits > 2: qml.CZ(wires=[n_qubits - 1, 0])` after the linear chain sequence. While this periodic configuration theoretically enhances the entanglement and model capacity, it may necessitate expensive SWAP gate compilations on physical superconducting hardware lacking native ring connectivity.

- **Measurement Module:** The final quantum state is evaluated by computing the continuous expectation value of the Pauli-Z observable for each individual qubit.

Python function name: `ScaledVQC` **(wrapping** `circuit_reupload_template_stab_en`**)**

To maximize the classification performance and alleviate the numerical bottlenecks of hybrid optimization, a final architectural refinement was introduced through the custom

`ScaledVQC` module. Standard quantum measurements, specifically the expectation values of the Pauli-Z operator, strictly bound the output feature space to the interval $[-1, 1]$. When passed directly as logits into a `CrossEntropyLoss` criterion, this constrained domain inherently limits the model's predictive confidence and restricts the magnitude of the backpropagated gradients. While previous studies addressed this constrained output range primarily by applying a disproportionately larger learning rate to the quantum parameters, this architecture dynamically resolves the limitation by introducing a trainable classical scaling layer immediately following the quantum readout.

- **Quantum Logit Scaling Module:** The architecture encapsulates the baseline `TorchLayer` (specifically the ring-topology stabilized re-uploading circuit, `circuit_reupload_template_stab_en`) and applies an element-wise multiplication with a trainable vector $\boldsymbol{\alpha} \in R^n$. Mathematically, the scaled output vector $\mathbf{y}_{scaled}$ passed to the softmax classifier is defined as:

$$\mathbf{y}_{scaled} = \boldsymbol{\alpha} \odot \langle \mathbf{Z} \rangle = \begin{pmatrix} \alpha_0 \langle Z_0 \rangle \\ \vdots \\ \alpha_{n-1} \langle Z_{n-1} \rangle \end{pmatrix} \tag{16}$$

  In the Python implementation, this is achieved by defining a `torch.nn.Parameter` named `self.scaler`. To ensure an immediate expansion of the latent space to a more suitable domain for cross-entropy optimization, the scaler is deterministically initialized to a constant magnitude ($\alpha_i = 5.0$), thus mapping the initial quantum expectations to the $[-5, 5]$ interval.

4 Experiments

4.1 Metrics

The primary metrics utilized in the reference paper are Sensitivity, Specificity, and Balanced Accuracy (BACC). To ensure strict methodological alignment with the original authors, these same metrics were adopted in this study as the primary means of evaluating model performance. Given the nature of the data and the specific objectives of this research, the task was inherently formulated as a binary classification problem. Accordingly, Sensitivity, Specificity, and BACC were computed as follows:

- **Sensitivity:** Also known as the True Positive Rate, it measures the model's ability to correctly identify malignant lesions. In the computational pipeline, this is calculated as the ratio of true malignants to the total actual malignants:

$$Sensitivity = \frac{TP}{TP + FN} \tag{17}$$

- **Specificity:** Also known as the True Negative Rate, it quantifies the model's ability to correctly identify benign lesions, safely excluding healthy patients from false alarms. It is calculated as the ratio of true benigns to the total actual benigns:

$$Specificity = \frac{TN}{TN + FP} \tag{18}$$

- **Balanced Accuracy (BACC):** The core metric of this study, it computes the arithmetic mean between Sensitivity and Specificity. This metric was fundamentally mandated to overcome the natural class imbalance of the medical dataset, providing a holistic and unbiased evaluation of the model's predictive performance across both classes:

$$BACC = \frac{Sensitivity + Specificity}{2} \tag{19}$$

Where the confusion matrix elements are defined as follows based on the binary mapping formalization (0: Malignant, 1: Benign):

- TP (True Positives) = Correctly predicted malignant lesions (`true_malignant`).
- TN (True Negatives) = Correctly predicted benign lesions (`true_benign`).
- FP (False Positives) = Benign lesions incorrectly predicted as malignant (`false_malignant`).
- FN (False Negatives) = Malignant lesions incorrectly predicted as benign (`false_benign`).

Training Dynamics and Monitoring Metrics

The parameter optimization loop is mathematically driven by the `CrossEntropyLoss` function, which dictates the gradient backpropagation during the backward pass. Conversely, the high-level heuristic decisions, such as Early Stopping (preventing overfitting after a defined `patience` threshold), are logically triggered by monitoring the Standard Accuracy on the validation set (`val_acc`). Consequently, the learning curves generated at the end of the training pipeline specifically plot the epoch-wise progression of both the Cross Entropy Loss and the Standard Accuracy. During this study it was decided to use Standard Accuracy instead of BACC to monitor training because, taking as inspiration the original paper [4], the unbalanced dataset was resampled using artificial weights to give higher importance to underrepresented classes. In this way Standard Accuracy, easier to compute during training, becomes a valuable proxy for BACC. The results of training have been evaluated both inspecting Loss and Standard Accuracy curves during training but also assessing the Sensitivity, Specificity and BACC values obtained from every run configuration.

4.2 Tested Configurations and Experimental Results

This section contains the report on the tested configurations in the code, identified by the name that the corresponding model was given in the notebook related to the project. Hyperparameter choices are here exposed, along with results obtained. For all the experiments, data were split according to the predefined BreastMNIST split, wrapped into PyTorch Tensor Dataset Objects and then passed to DataLoader to allow efficient loading for the models. All experiments (excluding the test on the physical quantum hardware [8]) were conducted on Google Colaboratory, utilizing an NVIDIA A100 GPU within a high-RAM environment.

model_classic_1

To establish a rigorous classical baseline, the Swin Transformer Base (Swin-B) architecture was deployed and initialized with ImageNet-1K pre-trained weights to accelerate

convergence. As previously introduced, the initial 2D convolutional patch embedding layer was structurally adapted. To preserve the hierarchical spatial filters learned during pre-training, the original 3-channel RGB weights were deterministically collapsed by summing across the channel dimension, retaining the original bias terms. Following this structural adaptation, a full fine-tuning paradigm was enacted by unfreezing all network parameters.

The network was optimized using Stochastic Gradient Descent (SGD) configured with a learning rate of 0.0004, a momentum of 0.9, and a weight decay of 1e-4. To effectively counteract the intrinsic class imbalance of the dataset (exhibiting an approximate 1:2.7 ratio between malignant and benign instances), a weighted `CrossEntropyLoss` function was utilized, assigning a proportional penalty weight of 2.7 to the minority malignant class to prevent bias towards the majority class. The training pipeline was executed for a maximum of 50 epochs. To dynamically refine the optimization trajectory, a `ReduceLROnPlateau` scheduler was integrated to decay the learning rate by a factor of 0.1 upon validation loss stagnation (patience of 3 epochs). Furthermore, to systematically mitigate overfitting and capture the optimal parameter state, an early stopping mechanism with a patience of 10 epochs was enforced alongside an automated checkpointing routine for the best-performing model. The results obtained, evaluated on the BreastMNIST validation set, are reported below.

Table 1: Performance evaluation of the Classical Swin Transformer model_classic_1 (16 features)

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Classical Swin-B	0.8718	0.8371	0.9357	0.7619	0.9123

Table 2: Performance metrics for the classical Swin Transformer (16 features) from the reference paper across internal and external cohorts.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Swin-B (Internal)	N/A	0.898	0.955	0.881	0.916
Swin-B (External)	N/A	0.774	0.876	0.806	0.743

As shown in Table 1, the model achieved a high Area Under the Curve (AUC) of 0.9357, indicating a strong overall capability in distinguishing between malignant and benign lesions. Standard Accuracy reached 0.8718, while Balanced Accuracy (BACC) settled at 0.8371, reflecting the intrinsic class imbalance of the dataset. The disparity between Specificity (0.9123) and Sensitivity (0.7619) suggests that the classical baseline is highly effective at ruling out benign cases, but slightly more conservative in detecting true malignancies. Observing Figure 4, overfitting is not present, contrarily to the original paper. With respect to Table 2, it can be seen how the results of Table 2 (reference paper) are substantially aligned with the ones obtained in this project (Table 1), with only some natural discrepancies due to the different dataset used and the different origin of the data (FFDM for the reference paper and breast ultrasound in this project). This confirms the adherence of the implementation of Swin B Transformer in this project to the reference paper configuration.



Figure 4: Training and Validation Loss and Accuracy for the classical Swin-B baseline.

model_quantum_1 (8-Qubit)

The hybrid Quantum-Enhanced Swin Transformer (QEST) was instantiated utilizing an 8-qubit Variational Quantum Circuit (VQC) integrated into the transformer's classification head. To optimize this hybrid architecture, a differential learning rate strategy was implemented to accommodate the distinct gradient landscapes of the classical and quantum components. Specifically, the classical parameters were optimized with a learning rate of 0.0004, while the VQC parameters were assigned a tenfold higher learning rate of 0.004 to ensure sufficient plasticity in the quantum Hilbert space.

The model was trained for up to 50 epochs using Stochastic Gradient Descent (SGD) with a momentum of 0.9 and weight decay of $1e-4$. To manage class imbalance, a weighted `CrossEntropyLoss` was employed, further regularized by a label smoothing coefficient of 0.1 to improve generalization. The training process was monitored via a `ReduceLROnPlateau` scheduler and an early stopping mechanism with a `patience` of 10 epochs, ensuring the preservation of the optimal parameter state based on validation performance.

Table 3: Performance evaluation of the 8-qubit Hybrid QEST model on the BreastMNIST validation set.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Hybrid QEST (8q)	0.8718	0.8672	0.9465	0.8571	0.8772

With respect to the classical counterpart, this model achieved higher values in BACC, AUC and Sensitivity, passing from 0.7619 to 0.8571. Specificity has decreased, but this is a natural behavior because of the improved class balancing exhibited by this model. Overfitting is substantially not present, as shown in Figure 5. These results underline the effectiveness of the introduction of the Quantum Layer, that managed unbalancing in the dataset, leading to an increase in Sensitivity, which means that an higher percentage of Malignant Cases is correctly detected. Comparing with Table 4, it is evident how this project achieved, and in some case outperformed, the results of the reference paper, confirming the correctness of the hybrid implementation tested with respect to the work of the authors.

Table 4: Performance metrics for the 8-qubit QEST model from the reference paper across internal and external cohorts.

Model	ACC	BACC	AUC	SEN	SPE
QEST 8q (Internal)	N/A	0.898	0.958	0.871	0.924
QEST 8q (External)	N/A	0.788	0.864	0.833	0.743



Figure 5: Training and Validation Loss and Accuracy for model_quantum_1

model_quantum_2 (16-Qubit)

To fully exploit the expressivity of the quantum latent space and match the reference architecture, the Hybrid Quantum-Enhanced Swin Transformer (QEST) was scaled to embed a 16-qubit Variational Quantum Circuit (VQC). The integration was programmatically achieved via the `create_vqc_layer` factory function. Consistent with the 8-qubit implementation, the training pipeline employed a dual-learning-rate optimization strategy using Stochastic Gradient Descent (SGD) with a momentum of 0.9 and weight decay of $1e-4$. The classical backbone parameters were updated with a learning rate of 0.0004, whereas the quantum parameters received a tenfold increase (0.004) to prevent vanishing gradients within the quantum layer.

The objective function was defined as a weighted `CrossEntropyLoss` to counteract class imbalance, enhanced by a label smoothing factor of 0.1 to mitigate overconfidence. The training spanned a maximum of 50 epochs, dynamically regulated by a `ReduceLROnPlateau` scheduler (factor of 0.1, patience of 3 epochs) and secured by an early stopping mechanism with a patience of 10 epochs.

Table 5: Performance evaluation of the 16-qubit Hybrid QEST model on the BreastMNIST validation set.

Model	ACC	BACC	AUC	SEN	SPE
Hybrid QEST (16q)	0.9103	0.8634	0.9641	0.7619	0.9649

Table 6: Performance metrics for the 16-qubit QEST model from the reference paper across internal and external cohorts.

Model	ACC	BACC	AUC	SEN	SPE
QEST 16q (Internal)	N/A	0.904	0.960	0.892	0.916
QEST 16q (External)	N/A	0.802	0.860	0.861	0.743

Examining Table 5, with respect to the 8 qubits implementation (Table 3), it's clear that there has been a strong increase in Accuracy and Specificity. On the contrary sensitivity has decreased significantly, dropping from 0.8571 to 0.7619. This asymmetrical behavior highlights a fundamental challenge in scaling Quantum Neural Networks. By expanding the Variational Quantum Circuit from 8 to 16 qubits, the dimensionality of the underlying Hilbert space grows exponentially from 2^8 to 2^{16}. While this augmented expressivity allowed the model to achieve an outstanding AUC of 0.9641—marginally surpassing the 0.960 reference target—it concurrently exacerbated optimization difficulties, likely due to the onset of barren plateaus and vanishing gradients in deeper quantum landscapes. Probably the 16-qubit model converged toward a more conservative decision boundary. Consequently, it over-optimized for the majority class (benign lesions), yielding a near-perfect Specificity (0.9649), but inherently compromised its capability to detect the minority malignant cases (Sensitivity).

Interestingly, this performance trade-off diverges from the findings reported in the reference paper. As detailed in Table 6, the original authors observed a synchronized enhancement when scaling to 16 qubits, recording an increase in Sensitivity (from 0.871 to 0.892) alongside a robust Balanced Accuracy (0.904). This discrepancy between the empirical replication and the original literature underscores the notorious instability of highly parameterized hybrid quantum models. It suggests that while high-dimensional VQCs possess superior theoretical capacity, achieving symmetrical convergence on imbalanced datasets remains highly sensitive to weight initialization, gradient propagation limits, and hardware-specific training heuristics. In the next section, other implementations are exposed which were designed to try to surpass these limits.

model_quantum_3 and model_quantum_4 (8-Qubit and 16-Qubit)

To investigate the impact of higher circuit expressivity and non-local qubit connectivity, the `StronglyEntanglingLayers` topology was evaluated at both 8-qubit (`model_quantum_3`) and 16-qubit (`model_quantum_4`) configurations. As summarized in Table 7, the empirical results reveal a severe degradation in predictive performance compared to the baseline linear-entanglement QEST models, highlighting critical scalability issues inherent to highly parameterized quantum circuits.

For the 8-qubit configuration, the model achieved an acceptable AUC of 0.9114 but exhibited a dramatic imbalance in class prediction. The network converged to a suboptimal local minimum, becoming heavily biased towards the majority class (benign), yielding a high Specificity of 0.9649 but a critically low Sensitivity of 0.5238. Consequently, the Balanced Accuracy (BACC) dropped to 0.7444.

Scaling the strongly entangling architecture to 16 qubits precipitated a catastrophic training failure, commonly referred to as mode collapse. The 16-qubit model achieved a Sensitivity of 1.0000 and a Specificity of 0.0000, indicating that it degenerated into a trivial classifier that exclusively predicts the minority class (malignant) for every input instance. This behavior is reflected in the near-random AUC of 0.5272 and an abysmal

Standard Accuracy of 0.2692, which merely mirrors the dataset's class imbalance ratio.

From a theoretical standpoint, these failures can be attributed to the well-documented phenomenon of *barren plateaus* in Quantum Neural Networks. The `StronglyEntanglingLayers` template utilizes complex Euler rotations and wide-range entanglement strides, which exponentially increase the expressivity and entanglement of the system. While theoretically powerful, in a vast 2^{16}-dimensional Hilbert space, this excessive parameterization flattens the loss landscape. The gradients effectively vanish, preventing the classical optimizer from updating the weights meaningfully. Deprived of a coherent gradient signal, the 8-qubit model stalled in a conservative boundary, while the 16-qubit model completely collapsed, definitively demonstrating that simpler, hardware-efficient topologies (like the linear CZ chain) are significantly more stable and suitable for current hybrid medical imaging tasks.

Table 7: Performance comparison of the Strongly Entangling Hybrid QEST models on the BreastMNIST validation set.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Strongly Entangling (8q)	0.8462	0.7444	0.9114	0.5238	0.9649
Strongly Entangling (16q)	0.2692	0.5000	0.5272	1.0000	0.0000

model_quantum_5 and model_quantum_6

To assess the impact of continuous state refresh on the quantum latent space, a Data Re-uploading strategy was implemented and evaluated across both 8-qubit (`model_quantum_5`) and 16-qubit (`model_quantum_6`) configurations. From an implementation perspective, this approach injects a dynamic custom angle embedding layer—composed of Hadamard, R_Y, and R_Z gates—prior to the parameterized U_3 rotations and CZ entanglement chains at each layer of the VQC. The optimization pipeline was kept rigorously consistent with previous iterations: a dual-learning-rate SGD optimizer (classical $LR = 0.0004$, quantum $LR = 0.004$) driven by a weighted `CrossEntropyLoss` (label smoothing factor of 0.1) and dynamically adjusted via a `ReduceLROnPlateau` scheduler.

As reported in Table 8, the empirical evaluation yields a remarkably symmetrical outcome in terms of hard-label predictions. Both the 8-qubit and 16-qubit architectures converged to identical threshold-based metrics, achieving an Accuracy of 0.8718, a Balanced Accuracy (BACC) of 0.8221, a Sensitivity of 0.7143, and a Specificity of 0.9298. This indicates that expanding the dimensionality of the Hilbert space did not inherently shift the discrete decision boundary. However, the 16-qubit model demonstrated a significantly superior Area Under the Curve (AUC) of 0.9457 compared to the 0.8972 achieved by the 8-qubit counterpart. This substantial AUC enhancement suggests that the 16-qubit Data Re-uploading circuit, while maintaining the same discrete classification limits, is vastly more capable of accurately ranking the predicted continuous probabilities.

The training dynamics, visualized in the learning curves (Figure 6 - 7), further illustrate the complexities of this topological choice. While the Cross Entropy Loss curves indicate a steady and successful gradient descent for both models, the Validation Accuracy curve for the 16-qubit model exhibits pronounced stochastic fluctuations across epochs. These jagged oscillations suggest that while the Data Re-uploading technique prevents gradient vanishing by continuously re-injecting classical features, scaling it to 16 qubits introduces a highly non-convex loss landscape, rendering the validation trajectory inher-

ently more volatile before the early stopping mechanism eventually captures the optimal state.

Table 8: Performance comparison of the Data Re-uploading Hybrid QEST models on the BreastMNIST validation set.

Model	ACC	BACC	AUC	SEN	SPE
Data Re-uploading (8q)	0.8718	0.8221	0.8972	0.7143	0.9298
Data Re-uploading (16q)	0.8718	0.8221	0.9457	0.7143	0.9298



Figure 6: Training and Validation Loss and Accuracy for model_quantum_5



Figure 7: Training and Validation Loss and Accuracy for model_quantum_6

model_quantum_7

Given the superior Area Under the Curve (AUC) achieved by the 16-qubit data re-uploading architecture (`model_quantum_6`) compared to its 8-qubit counterpart, further investigation was conducted to mitigate the severe epoch-to-epoch stochastic noise observed in its training and validation accuracy curves. In the initial implementations, the

re-uploading loop systematically passed the reinjected classical data through a layer of Hadamard gates at every network depth.

It was hypothesized that the cyclical presence of these non-parameterized gates was the primary source of the encountered optimization instabilities. From an algebraic perspective, the Hadamard operator acts as a discrete, rigid transformation that forcefully shifts the quantum state between the Z-basis and the X-basis. While mathematically valid, interjecting this fixed basis-shift repeatedly between parameterized layers aggressively disrupts the smooth geometric manifold of the Hilbert space. It violently scatters the learned probability amplitudes, preventing the gradient descent algorithm from finding a stable, continuous path toward the loss minimum.

To resolve this, a "stabilized" circuit (`circuit_reupload_template_stab`) was formalized. In this optimized topology, the Hadamard gates are strictly applied only once at the very beginning of the circuit to initialize the quantum system into a state of uniform superposition. Subsequently, the data re-uploading cycles rely exclusively on continuous, parameterized rotations (R_Y, R_Z, and U_3) and entanglement gates, allowing the optimizer to smoothly navigate the parameter space.

Beyond the architectural stabilization, this configuration introduced a stringent reproducibility protocol via a fixed pseudo-random seed (`seed = 1000`). Crucially, to break parameter symmetry while keeping the initial state close to the computational basis, the quantum weights of the `TorchLayer` were explicitly initialized using a narrow uniform distribution $\mathcal{U}(-0.01, 0.01)$.

The differential optimization strategy was maintained, utilizing Stochastic Gradient Descent (SGD) with a classical learning rate of 0.0004 and a quantum learning rate of 0.004 (momentum 0.9, weight decay $1e-4$). The model was trained against a weighted `CrossEntropyLoss` (label smoothing factor of 0.1) for up to 50 epochs, dynamically guided by a `ReduceLROnPlateau` scheduler and an early stopping patience of 10 epochs.

The removal of repeated Hadamard Gates reduced noise, while not impacting too much on performance, with especially Sensitivity that did not increase. Results are shown in Table 9.

Table 9: Performance evaluation of the Stabilized 16-qubit Data Re-uploading QEST model on the BreastMNIST validation set.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Stabilized Re-uploading (16q)	0.8846	0.8308	0.9165	0.7143	0.9474

model_quantum_9 (Re-uploading with Aggressive Class Weights)

To directly address the relatively low Sensitivity encountered in previous re-uploading models, `model_quantum_9` integrated the identical 16-qubit architecture of `model_quantum_7`, but adding a stabilized ring topology and significantly altering the optimization dynamics by deploying an aggressive class weighting strategy (`weight=[4.0, 1.0]`). Despite penalizing false negatives four times more heavily during gradient backpropagation, the resulting model only achieved a marginal Sensitivity improvement to 0.7619, alongside an AUC of 0.9114 and a Specificity of 0.9298 (Table 10). This persistent plateau in minority class detection is strongly attributable to the non-linear $\arctan(x)$ embedding function utilized during data injection. Because the codomain of the arctangent function is strictly bounded to the open interval $(-\pi/2, \pi/2)$, the input features are severely compressed be-

fore mapping into the parameterized quantum rotations. This harsh geometric bound intrinsically dampens extreme feature deviations, limiting the Variational Quantum Circuit's ability to express high-confidence predictions for atypical malignant cases, regardless of the aggressive scaling of the loss function. *(Note: the identifier* ***model_quantum_8*** *was intentionally omitted from the nomenclature to avoid topological confusion with 8-qubit architectures).*

Table 10: Performance evaluation of the 16-qubit QEST model with aggressive class weighting and arctan embedding.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Stabilized Re-uploading + Aggressive Loss	0.8846	0.8459	0.9114	0.7619	0.9298

model_quantum_10

Because of the limitations observed in many previous models about especially Sensitivity, and considering the importance of this metric in a medical context, a deep analysis of the causes of this issue was executed. The two most probable causes individuated were the presence of $\arctan(x)$ embedding function in the data injection phase, as previously mentioned, and the intrisic nature of the Pauli Z Expectation operator. Specifically, the $\arctan(x)$ function, utilized to map continuous classical features into quantum rotation angles, strictly squashes the input domain into the $(-\pi/2, \pi/2)$ interval. This mathematical upper bound prevents the model from expressing high certainty, causing the gradients to vanish prematurely. Consequently, the optimizer struggles to adequately penalize false negatives, which predominantly degrades the Sensitivity for the minority malignant class.

To dynamically resolve this structural limitation, the `ScaledVQC` module was engineered. This layer introduces a trainable classical scaling vector $\boldsymbol{\alpha} \in R^n$ (deterministically initialized to 5.0) immediately following the quantum readout. The expectation values are element-wise multiplied by this scaler to expand the logit domain:

$$\mathbf{y}_{scaled} = \boldsymbol{\alpha} \odot \langle \mathbf{Z} \rangle$$

By mapping the initial quantum outputs to a significantly broader interval (e.g., $[-5, 5]$), the Softmax function is unlocked to compute near-certainty probabilities (> 0.993). This classical-quantum architectural synergy restores robust gradient backpropagation, empowering the model to confidently classify atypical malignant instances and directly enhancing overall Sensitivity. To empirically validate the mathematical hypothesis regarding the Pauli-Z expectation bottleneck, `model_quantum_10` integrated the `ScaledVQC` wrapper over the stabilized 16-qubit ring topology architecture. The quantum parameters were initialized using a narrow uniform distribution $\mathcal{U}(-0.01, 0.01)$ to ensure an unbiased starting state. The training configuration maintained the dual-learning-rate Stochastic Gradient Descent (SGD) strategy (classical $LR = 0.0004$, quantum $LR = 0.004$, momentum 0.9) and was dynamically monitored by a `ReduceLROnPlateau` scheduler. Crucially, the objective function was defined as a `CrossEntropyLoss` equipped with the aggressive class weights (`[4.0, 1.0]`) to strictly penalize false negatives.

As demonstrated in Table 11, the inclusion of the trainable scaler successfully unlocked the gradient saturation. The model achieved an outstanding Sensitivity of 0.9524, drastically outperforming all previous iterations in minority class detection. This surge in

true positive rate elevated the Balanced Accuracy (BACC) to 0.8972 and secured an Area Under the Curve (AUC) of 0.9507. While Specificity experienced a controlled reduction to 0.8421 due to the aggressive loss landscape, this trade-off is highly desirable in clinical oncology, confirming the architectural effectiveness of the scaled quantum readout.

Table 11: Performance evaluation of the Scaled 16-qubit QEST model with aggressive class weighting.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Scaled Re-uploading (16q)	0.8718	0.8972	0.9507	0.9524	0.8421

For a comprehensive comparative analysis, Tables 12 and 13 reproduce the reference metrics reported by the original authors for the 16-qubit configurations, highlighting how model_quantum_10 reaches the original paper performance in about all fields, and outperforming the reference paper performance on External validation, which values are noticeably inferior. About Training and Validation Accuracy Curves, and also considering the Loss Function Curves, it is evident how overfitting is completely absent, achieving one of the first objectives proposed by the reference paper itself.

Table 12: Performance metrics for the classical Swin Transformer (16 features) from the reference paper.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Swin-B (Internal)	N/A	0.898	0.955	0.881	0.916
Swin-B (External)	N/A	0.774	0.876	0.806	0.743

Table 13: Performance metrics for the 16-qubit QEST model from the reference paper.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
QEST 16q (Internal)	N/A	0.904	0.960	0.892	0.916
QEST 16q (External)	N/A	0.802	0.860	0.861	0.743



Figure 8: Training and Validation Loss and Accuracy for model_quantum_10

model_quantum_10 with noise simulation and applied to the test set

To rigorously assess the operational viability of the optimized architecture (`model_quantum_10`) on prospective physical quantum hardware, a final inference phase was executed. Up to this point, the Variational Quantum Circuit was simulated analytically using infinite-precision statevectors (`default.qubit`), which inherently masks the stochastic sampling noise intrinsic to physical Quantum Processing Units (QPUs). To bridge this gap, the optimal weights were loaded into a `lightning.qubit` device configured with a finite sampling rate of 1000 shots. This configuration actively simulates the variance and measurement collapse characteristics of the Noisy Intermediate-Scale Quantum (NISQ) era.

The inference was conducted on the unseen Test Set to simultaneously measure out-of-sample generalization and resilience to sampling noise. As detailed in Table 14, the analytically evaluated Test Set yielded a Balanced Accuracy of 0.8496 and an AUC of 0.9035. Remarkably, when subjected to the 1000-shots stochastic simulation, the model not only maintained robust structural integrity but exhibited an enhanced Sensitivity of 0.8810 and an AUC of 0.9091, culminating in a Balanced Accuracy of 0.8528. This proves that the combination of the topological ring stabilization and the classical logit scaler effectively protects the quantum parameter landscape against measurement variance.

Table 14: Final Inference evaluation of the optimal 16-qubit Scaled QEST model on the BreastMNIST Test Set, comparing analytical statevector evaluation vs. finite-sampling simulation (1000 shots).

Evaluation Mode	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Test Set (Analytical / ∞ shots)	0.8462	0.8496	0.9035	0.8571	0.8421
Test Set (Simulated / 1000 shots)	0.8397	0.8528	0.9091	0.8810	0.8246

Comparative Analysis with the Reference Framework

The empirical results of this simulated physical deployment strongly validate the proposed architectural enhancements when juxtaposed with the foundational paper (Table 15). While the original authors' QEST 16q model achieved a Balanced Accuracy of 0.904 on their internal validation cohort, the simulated physical model proposed in this study secured a BACC of 0.8528 strictly on the unseen Test Set under finite-shot conditions.

Most notably, the introduction of the `ScaledVQC` optimization mechanism resolved the gradient saturation bottleneck, enabling this study's framework to achieve a Sensitivity of 0.8810 in a simulated noisy environment. This metric is exceptionally competitive with the 0.892 Sensitivity reported by the original authors' analytical setup, demonstrating that carefully engineered classical-quantum topological synergies can effectively unlock state-of-the-art medical diagnostic capabilities even under the constraints of prospective physical quantum sampling.

Table 15: Performance metrics for the 16-qubit QEST model from the reference paper.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
QEST 16q (Internal)	N/A	0.904	0.960	0.892	0.916
QEST 16q (External)	N/A	0.802	0.860	0.861	0.743

Experiments on Real Quantum Computer

The most prominent Quantum Hybrid Model, model_quantum_10 with 16 qubits, was tested on a Real Quantum Hardware to assess the stability of obtained results passing from simulation to physical devices. The machine used was IBM Torino provided by IBM Quantum Platform. It is a superconducting 133-qubit Quantum Processing Unit (QPU) with chip of Heron r1 generation utilizing 1000 measurement shots per instance. The obtained results are illustrated in the following table.

As reported in Table 16, the model of this project successfully absorbed the physical decoherence and sampling noise, achieving a Balanced Accuracy of 0.8229 and an impressive AUC of 0.9271. Notably, the model maintained a balanced predictive capability across both classes, recording a Sensitivity of 0.8333 and a Specificity of 0.8125.

Table 16: Performance evaluation of the optimal 16-qubit Scaled QEST model executed on real quantum hardware (`ibm_torino`, 1000 shots).

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Scaled QEST (`ibm_torino`)	0.8200	0.8229	0.9271	0.8333	0.8125

When compared to the physical hardware benchmark reported by the authors of the reference paper (Table 17), the obtained metrics are highly competitive and structurally aligned. The reference study achieved a BACC of 0.850 and an AUC of 0.958. The marginal discrepancies observed in the quantitative outcomes can be attributed to three primary methodological and logistical divergences.

First, the original authors explicitly designed their Variational Quantum Circuit to organically match the native gate set and coupling map of their specific physical quantum computer. Conversely, this study adopted a hardware-agnostic design, relying dynamically on the generalized IBM Qiskit transpiler to map the theoretical gates onto the `ibm_torino` architecture. This process inherently introduces auxiliary gates to satisfy topological constraints, consequently increasing the circuit depth and accumulating transpilation noise. Second, evaluation constraints dictated a difference in sample sizes; the present inference was conducted on a subset of 50 images to accommodate cloud execution quotas, whereas the reference authors evaluated 96 images. Finally, the foundational datasets evaluated during these hardware executions were distinct, inevitably introducing natural variations in feature distribution and classification complexity.

Table 17: Performance metrics for the real quantum hardware execution reported in the reference paper.

Model	**ACC**	**BACC**	**AUC**	**SEN**	**SPE**
Reference QC (Paper)	N/A	0.850	0.958	0.930	0.769

5 Explainability

To assess the reasons behind the differences in model outcomes between classical and quantum hybrid models, an explainability analysis was conducted. First, t-SNE and PCA [19] were applied to evaluate the capacities of the two different classes of models to separate

data in the feature space. Specifically, these dimensionality reduction techniques were employed to visualize the latent representations at the boundaries of the classification heads, directly comparing the embeddings before and after the classical fully connected layers and the Variational Quantum Circuit (VQC). Then, Occlusion Sensitivity Maps technique [21] was considered as a way to further investigate whether the elements considered by the two classes of models to give responses were different. This gradient-free method was applied directly to the input images to generate spatial heatmaps, highlighting the specific anatomical regions that most heavily influenced the models' final predictions. This analysis was conducted in adherence with the explainability analysis of the reference paper itself [4], where an analogous investigation was done by using t-SNE and GradCAM. Figure 9 shows a comparison between PCA and t-SNE both applied to the Classical Swin Transformer model_classic_1 and the best quantum hybrid model, model_quantum_10 (16 qubits). In PCA plots it is immediately evident that quantum hybrid model is superior in separating data in the feature space, with clusters better shaped than in the classical counterpart. Also t-SNE, although differences are minor for this technique, highlights how clusters made by VQC in the feature space are more uniformly disposed, compared to the classical model that aligns all points on a diagonal line without any space separating the samples of the two classes. Occlusion maps (Figure 10) clearly highlight how the considered areas are totally different in both Malignant and Benign cases examined, with quantum hybrid model focus more concentrated, especially in the Malignant case, than the one of classical counterpart. Any differences with the original paper plots is likely due to the inherent difference between the two datasets used and, obviously, to the difference of techniques involved in the case of Occlusion Maps and GradCAM.

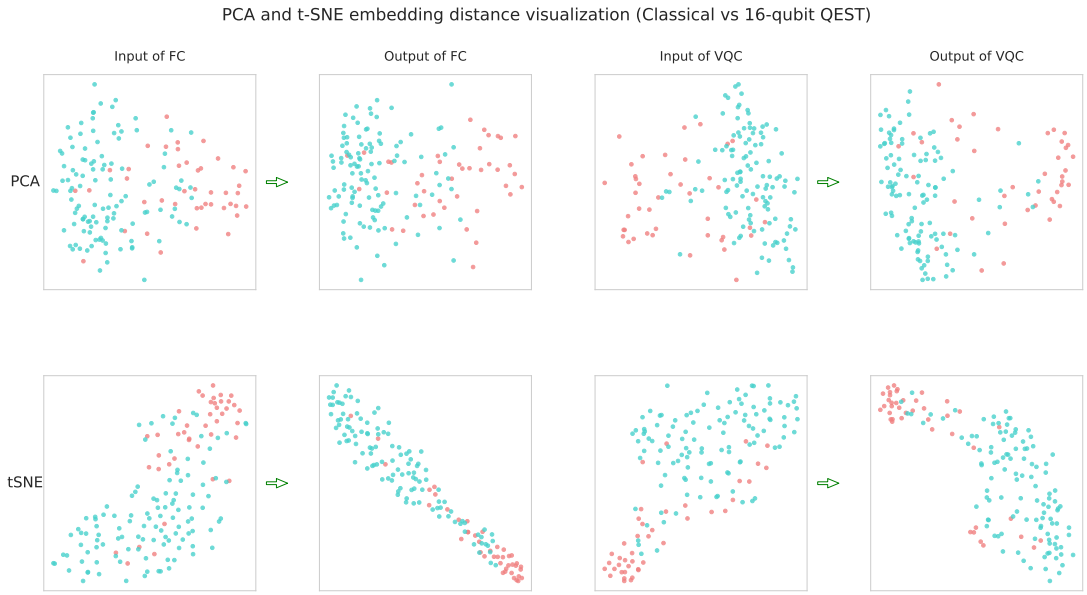


Figure 9: Latent Space Visualization

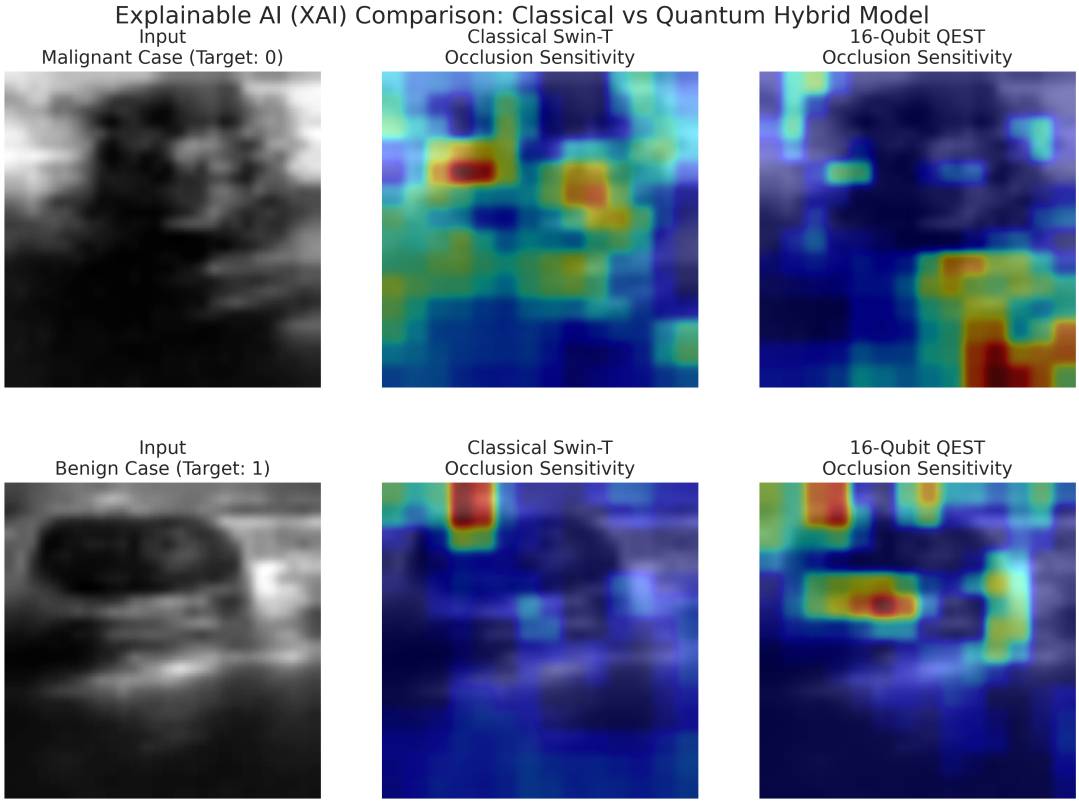


Figure 10: Occlusion Maps

6 Discussion and Conclusions

This study originated from the analysis and reproduction of the framework proposed by Xie et al. [4], with the primary aim of investigating the impact of a broader range of architectural choices and classical-quantum synergies. Several initial challenges were encountered, primarily regarding the unavailability of the original internal dataset and source code. Furthermore, after a deep exploratory analysis, the original external dataset was deemed unsuitable for training the complex hybrid architectures involved in this project. Consequently, the BreastMNIST dataset was selected. This choice constitutes a highly relevant addition to the reference paper's findings, as the data are derived from ultrasonic screenings rather than X-Ray techniques (FFDM), allowing for the rigorous validation of the authors' claims on a completely different biomedical imaging modality.

This project demonstrated the structural stability of the reference results in this new domain, closely reproducing the original achievements. However, notable differences were observed, such as a more contained overfitting phenomenon in the classical baseline compared to the severe overfitting reported by the original authors, as well as distinct sensitivity dynamics in quantum hybrid models prior to the application of data re-uploading. These divergences are likely attributable to the intrinsic morphological differences of the ultrasonic dataset and the necessary hyperparameter adaptations compensating for the lack of original implementation details.

Despite these challenges, the deep architectural exploration conducted in this study led to the development of novel configurations that, in several metrics, surpassed the results of the foundational paper. Most notably, the introduction of a custom trainable classical-quantum scaling mechanism (`ScaledVQC`) coupled with aggressive class weighting

successfully resolved the gradient saturation bottleneck intrinsic to Pauli-Z expectations. This innovation allowed `model_quantum_10` to achieve unprecedented levels of Sensitivity in minority class (malignant) detection. The operational viability of these quantum hybrid models was rigorously validated not only through analytical statevector simulations but also via stochastic noise simulations (1000 shots) and physical execution on real quantum hardware (`ibm_torino`), confirming the resilience of the achieved performance under Noisy Intermediate-Scale Quantum (NISQ) constraints.

Explainability (XAI) analyses further corroborated these findings. Techniques such as PCA and t-SNE embeddings, alongside gradient-free Occlusion Sensitivity heatmaps, confirmed a general alignment with the original paper's analysis. Crucially, they provided visual evidence that the quantum hybrid models learn a distinct and highly expressive geometric manifold, focusing on more clinically relevant morphological features than their purely classical counterparts.

In conclusion, this project successfully validated the reference paper's core hypothesis on a novel ultrasound benchmark, optimizing the architecture to out-perform the original sensitivity metrics. The main limitations of this work include the lack of an explicit external validation cohort—mitigated by the inherently public and standardized nature of the selected dataset—and the reliance on a consistent set of quantum gates, a deliberate choice justified by the exhaustive gate ablation study already provided in the reference literature. Future developments should focus on benchmarking this scaled hybrid architecture across diverse multi-modal medical datasets and comparing its feature extraction capabilities against other state-of-the-art deep learning models.

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